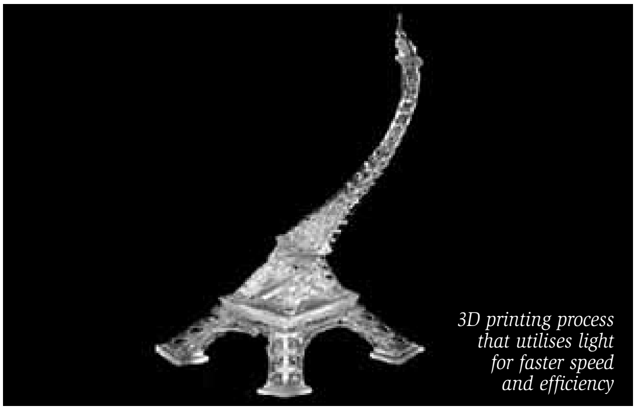
Tech News

Technology Updates

Light based 3D printing improves efficiency

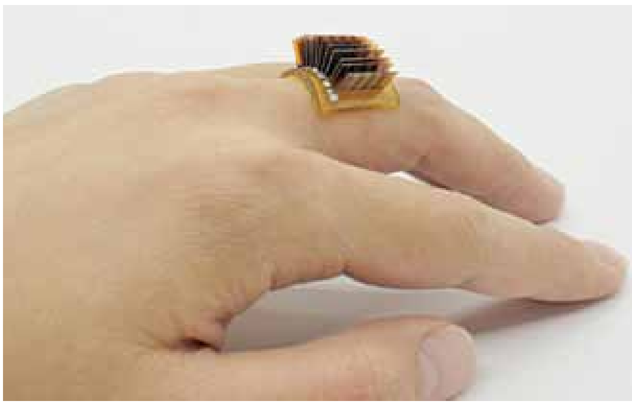
Conventional 3D printing processes create a digital design model that has printed layers assembled upwards like a cake, which requires more effort depending on the complexity of a design. Now, engineers at the Northwestern University have demonstrated that light can improve 3D printing speed and precision.

With the combination of a high-precision robot arm, there is more freedom to move each layer as the structure is built. By shining light on a liquid polymer, a crosslinking or polymerisation process occurs, converting liquid to solid and enabling continuous printing of up to 4,000 layers in approximately two minutes. This 'on-the-fly' feature enables the printing of more complicated structures and significantly improves manufacturing flexibility.



3D printing process that utilises light for faster speed and efficiency

Device that uses body heat to power wearables



A thermoelectric wearable device worn as a ring (Credit: www.colorado.edu)

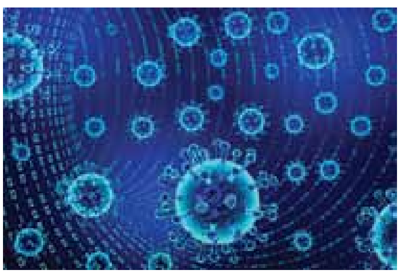
Researchers at CU Boulder have developed a device that is stretchy enough to be worn as a ring, a bracelet, or any other accessory. It taps into a person's natural body heat through thermoelectric generators to convert it into electricity. The device also has a unique ability to heal itself when damaged and is fully recyclable.

The device consists of a base made out of a stretchy material called polyimine to which a series of thin thermoelectric chips are connected. The final product looks like a cross between a plastic bracelet and a miniature computer motherboard or may be a techy diamond ring. The device generates about one volt for every square centimetre of skin space, which is less voltage per area than what most existing batteries provide but still enough to power electronics like watches or fitness trackers.

AI helps curb Covid-19 mutations

To counter emerging coronavirus strains, researchers at the USC Viterbi School of Engineering have devised a new method using artificial intelligence. It can quicken vaccine analysis and thus determine the best potential preventive medical therapy.

The machine-learning model accomplishes vaccine design cycles in a matter of seconds and minutes without compromising safety, giving humans a big advantage over the evolving contagion. Even if SARS-CoV-2 becomes uncontrollable by current vaccines, or if new vaccines are needed to deal with other emerging viruses, the method can be used to design other preventive mechanisms quickly.



The USC Viterbi machine-learning model can accomplish vaccine design cycles that once took months in a matter of minutes (Credit: https://news.usc.edu)